

FIELDWORKS MindLAB

FOUNDATIONAL ORGANIZATIONAL PROSPECTUS

FIELDWORKS MINDLAB

A founder-led laboratory for recursive field research, computational instrumentation, adaptive fabrication, and public scientific imagination

RESEARCH · INSTRUMENTATION · FABRICATION · PUBLIC INQUIRY

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Founder and Principal Investigator

Foundational Prospectus — Version 1.0

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PARTNERSHIP EDITION

Working research and organizational document. Patent-sensitive technical details require controlled circulation.

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This prospectus is designed as a modular system. Each numbered section can be extracted with its local references and contact block for a targeted audience.

Founder's Letter

My path to Fieldworks MindLAB has not been linear. It has moved through production graphic design, restaurant service, CNC subtractive fabrication in an architectural design studio, art-department work for music video production, tattooing, large-scale medicinal cannabis cultivation, music, engineering, and independent technical investigation. The common thread has been creation: taking something that begins as an idea, image, pattern, sound, or question and working through the process of making it real.

That drive led me toward engineering and physics. In August 2026, I graduated from ECPI University with a Bachelor of Science in Electrical Systems Engineering Technology and a focus in Mechatronics. Through electronics, embedded systems, automation, controls, mechanical design, robotics, machine learning, human-machine interfaces, and my senior project, I have learned how quickly even a small physical system becomes complex. Sensors introduce uncertainty. Feedback can stabilize or destabilize. Signals must be interpreted through noise. Design requires both mathematical reasoning and physical intuition.

My senior work combines embedded sensing, real-time data interpretation, visual feedback, and sound synthesis. What began as an engineering design challenge became a way to make invisible or abstract behavior perceptible. Alongside that work, I pursued procedural modeling, topology, recursive generative systems, and field-based design. Those investigations grew into the Cosmosis RDFT Suite and the broader research agenda gathered here.

After graduation, my direction is to deepen the mathematical and theoretical foundation necessary to approach these questions with greater rigor through advanced study, collaboration, and research. Fieldworks MindLAB is not a declaration that the difficult questions are already solved. It is an instrument for pursuing them with seriousness: building systems, measuring behavior, learning from failure, and creating partnerships between engineering practice, foundational physics, materials, computation, art, and public inquiry.

I invite collaborators who are willing to engage the work in that spirit—with curiosity, practical discipline, and the freedom to challenge its assumptions. My goal is to build a bridge between imagination and physical reality, and to make that bridge strong enough for other people to cross.

Brian Fahey
Founder and Principal Investigator
Fieldworks MindLAB

01 Executive Prospectus

Fieldworks MindLAB is an emerging founder-led research and creative-technology organization established by Brian Fahey to turn difficult questions about scale, topology, dynamics, material formation, perception, and computation into working instruments and testable programs. Its current foundation is the Cosmosis RDFT Suite: a public, experimental software environment combining generative fields, audio synthesis, environmental data, topology visualization, printable geometry, and measurement tools. The organization is seeking research partners, institutional hosts, equipment access, technical collaborators, and grant or seed support to move from a broad independent research platform toward disciplined, reproducible programs with defined milestones.

CORE PROPOSITION Fieldworks MindLAB builds instruments that make abstract field relationships observable, manipulable, audible, and fabricable. The near-term value is not a claim to have completed a new fundamental physics. It is the creation of an unusual experimental and design environment in which topology, recursive dynamics, environmental inputs, and material constraints can be explored together.

The MindLAB portfolio spans three mutually reinforcing modes. The first is theoretical: Recursive Differentiation and Recursive Differential Field Theory (RDFT) provide a speculative language for reasoning about resolution, nested structure, energy, and emergence. The second is instrumental: Cosmosis Navigator, the Visual Observatory, the Holo Drive synth, and the Hopf visualization kit turn that language into code, controls, logs, images, sound, and geometric probes. The third is translational: Surface Foundry, adaptive construction concepts, distributed sensing, and Cathode Ray Epitaxy ask whether field-informed design can produce useful structures or research methods.

This prospectus deliberately separates those modes. Established mathematics and physics are cited as context. Founder-developed models are identified as original frameworks. Classical measurements are not presented as direct observations of quantum systems. Simulations are not presented as material demonstrations. Proposed devices are accompanied by validation gates. This separation is an organizational strength: it gives partners a clear route to engage the work without having to endorse every long-range interpretation.

What Fieldworks MindLAB is asking for

- A host or collaborating institution capable of supporting rigorous technical review, experiment design, and publication.
- Access to fabrication, sensing, microscopy, vacuum, robotics, or advanced visualization facilities through structured partnerships.

- Co-investigators in applied mathematics, nonlinear dynamics, topology, materials science, controls, surface science, and science communication.
- Seed support for software hardening, reproducible datasets, benchmark experiments, and targeted prototypes.
- Grant-development partnerships that match each project to an appropriate readiness level and funding mechanism.
- Public-facing collaborators who can translate the audiovisual and fabrication work into STEAM learning, exhibitions, workshops, and community programs.

A portfolio rather than a single bet

Program	Current asset	Near-term proof	Partner value
Cosmosis RDFT Suite	Public multi-application codebase and working instruments	Reproducible releases, benchmark datasets, documented protocols	A flexible research and visualization testbed
Holonomy and topology	Classical closed-loop phase measurements across repeated sessions	Independent replication and controls	A tractable geometry experiment with clear falsification tests
Surface Foundry	Generative printable geometry and material-profile workflows	Fabricate and mechanically test a controlled specimen series	Design-to-fabrication research with terrestrial and off-world relevance
Cathode Ray Epitaxy	Formal engineering concept and staged simulation roadmap	Beam-source reaction-diffusion model and benchtop deposition study	A high-risk, high-upside maskless fabrication program
Voice of a Lyre	Distributed sensing architecture and conceptual white paper	Noise-floor model and small sensor-array prototype	Novel biomimetic sensing research without requiring exotic claims
Public MindLAB	Audio, visual, geometric, and environmental interfaces	Pilot workshops, installations, and learning evaluation	A compelling bridge between research, arts, and technical education

02 Identity, Mission, and Founder

Mission

Fieldworks MindLAB exists to develop and share instruments for exploring how patterns emerge across scales—mathematically, computationally, materially, sonically, visually, and experientially. It treats scientific inquiry and creative practice as complementary methods: measurement constrains the work; visualization and sound make structures perceptible; fabrication forces the models to confront material reality; and public engagement makes the process legible beyond a narrow specialist community.

Organizational character

The MindLAB is presently best understood as an independent research initiative moving toward a durable organizational form. This prospectus does not assume a legal status that has not yet been established. Different programs may ultimately require different vehicles: a research company for

sponsored development and IP, a nonprofit or fiscal partner for public education and arts programming, and formal subaward or collaboration agreements for university-led research.

Founder and Principal Investigator

Brian Fahey earned a Bachelor of Science in Electrical Systems Engineering Technology with a focus in Mechatronics from ECPI University in August 2026. He is an independent researcher, instrument builder, and creative technologist working across electronics, embedded systems, controls, robotics, machine learning, generative audio-visual design, nonlinear dynamics, topology, and fabrication. His distinctive contribution is systems integration: he has assembled a body of theory, code, interfaces, logging tools, analysis workflows, and project concepts that repeatedly translate abstract relationships into something a person can operate and observe.

Fahey's professional background includes production graphic design, restaurant service, CNC subtractive fabrication in a private architectural studio, art-department work for music video production, tattooing, and data-informed medicinal cannabis cultivation in a large-scale facility. These settings developed practical knowledge of materials, tools, process control, production metrics, communication, and the relationship between concept and execution. His current academic and independent interests include materials science, photonics and optics, quantum systems, magnetohydrodynamics, plasma propulsion, aerospace innovation, field behavior, and emergent complexity.

Fahey's role as Principal Investigator is not based on an assertion of settled authority across every discipline represented in the portfolio. It is based on authorship of the organizing framework, continuous development of the instrument suite, ownership of the research agenda, and responsibility for defining testable next steps. The partnership model is therefore intentionally collaborative: domain experts are invited not merely to endorse the work, but to challenge it, isolate what is valid, redesign what is weak, and help convert promising ideas into credible research programs.

FOUNDER POSITIONING Brian Fahey should be presented as Founder and Principal Investigator of Fieldworks MindLAB—originator of the research program and builder of its core instruments—while collaborators receive explicit credit for domain expertise, validation, facilities, data, and co-authored results.

Operating values

- Build before claiming: turn ideas into code, apparatus, protocols, or artifacts.
- Label epistemic status: distinguish established theory, analogy, measurement, simulation, and speculation.
- Invite falsification: define outcomes that would weaken or reject a proposed interpretation.
- Preserve reproducibility: publish source, versions, datasets, and procedures when IP and participant protections allow.
- Keep the work cross-disciplinary without erasing disciplinary standards.
- Use aesthetic experience to widen access, not to substitute for evidence.
- Treat safety, environmental responsibility, accessibility, and responsible communication as design requirements.

03 The Fieldworks Research Thesis

Across the portfolio, Fieldworks MindLAB asks a recurring question: what becomes possible when a system is described not only by objects and components, but by the relationships that make distinctions stable across scale? RDFT is Fahey's proposed language for that question. It treats resolution as an active structural variable and recursion as a mechanism through which one layer constrains or differentiates another.

In its strongest form, the framework makes claims about physics, gravity, energy, information, and consciousness. Those claims require substantial independent work and are not prerequisites for the applied program. A narrower and immediately useful interpretation treats RDFT as a generative modeling and control framework: a way to represent nested scales, drive fields, trace state transitions, explore attractors, and map high-dimensional dynamics into observable outputs.

Four layers of the thesis

Layer	Question	Current status	Research posture
Mathematical	Can recursive resolution fields be defined consistently and related to known scale-dependent formalisms?	Founder-developed equations and working references	Formalize, dimensional-check, compare, and invite critique
Computational	Do the equations and mappings produce stable, informative, reproducible dynamics?	Implemented across multiple applications	Benchmark, test, document, and release
Experimental analog	Can classical instruments exhibit controlled geometric, topological, or nonlinear signatures?	Initial holonomy and state-data studies	Replicate with controls and external review
Physical interpretation	Do any observed structures reveal new physical law or only useful analogies?	Open and speculative	Hold claims narrowly; design discriminating experiments

Why topology and holonomy recur

Topology offers a language for features that persist under continuous deformation. Holonomy asks what change remains after a state is transported around a closed loop. These ideas are established in geometry and physics, from Berry phase and Chern invariants to holonomic quantum control. Within Cosmosis they also become design tools: loops can be programmed, orientation can be reversed, enclosed area can be varied, and the return phase can be logged.

The attached Nielsen preprint places complex Hopf fibrations, nested shells, Beltrami operators, holonomy, and torsion at the center of an ambitious gauge–gravity unification framework. It is highly relevant to the conceptual development lineage of Fieldworks MindLAB, especially where Cosmosis uses Hopf maps, winding, Beltrami alignment, and geometric phase. It remains a preprint, and its broad physical deductions are treated here as a comparative research program, not as established validation of RDFT.

A disciplined synthesis

WORKING SYNTHESIS RDFT supplies a founder-developed language of recursive resolution. Hopf and bundle geometry supply established mathematical structures. Floquet and adiabatic theory supply controlled-drive concepts. Modern Hopfield networks supply a useful energy-based language for memory and attractor retrieval. Fieldworks MindLAB's research task is to make the interfaces among these domains explicit and testable.

04 Portfolio of Platforms and Programs

Cosmosis Navigator

Cosmosis Navigator is the mature performance and research instrument in the public suite. It combines generative field dynamics, Chladni-like pattern formation, synthesis control, environmental data bridges, quantum- and holonomy-themed panels, and SuperCollider routing. Its immediate research value is as a real-time state machine: parameters can be driven, logged, frozen, replayed, and mapped to multiple sensory channels.

Cosmosis Visual Observatory and Surface Foundry

The Visual Observatory separates visual and geometric exploration from the full audio instrument. It includes field atlases, a topology lab, material targets, relief and STL workflows, call-sheet generation, and constrained primitive domains. Surface Foundry is the translational bridge: a field configuration becomes a fabricable mesh, and a material profile becomes a set of parameters and process instructions.

Holo Drive RDFT/EG Synth

Holo Drive extends the instrument family through Floquet-style shaping, environmental bridges, lifecycle behaviors, and audio synthesis. Its value to the prospectus is not simply another application. It demonstrates portability of the framework across interfaces and the possibility of comparative experiments using shared state variables under different mappings.

RDFT Hopf Visualization Kit

The Hopf kit uses Python/PyGFX and optional TouchDesigner workflows to visualize Hopf, PCA, Berry, winding, and manifold portraits from logged RDFT runs. It is the clearest candidate for becoming an independent research tool: it can ingest standardized datasets, compare runs, expose uncertainty, and serve as a visual companion to formal analysis rather than as a purely aesthetic output.

BasinNET and energy-based learning

BasinNET is presently deferred from the public repository pending cleanup. Its conceptual role is nonetheless important. Modern Hopfield networks show that associative memory, energy minimization, and attention can be described within a shared mathematical form. A future BasinNET program can use that lineage to frame attractor basins, learned prototypes, and state retrieval without prematurely claiming a new theory of intelligence.

Translational program map

Domain	Platform function	Candidate outcome
Research software	Field simulation, logging, analysis, visualization	Versioned open research toolkit
Audio and performance	Audible mapping of dynamic structure	Concert, installation, research instrument
Advanced fabrication	Topology generation and process translation	Test coupons, structures, call sheets, STL outputs
Semiconductor research	Beam-path and growth-front modeling	CRE simulation and deposition prototype
Distributed sensing	Coherent array and noise-assisted detection concepts	Voice of a Lyre bench array
Public learning	Hands-on encounters with fields, feedback, and form	Workshops, exhibits, curriculum, evaluation data

05 Evidence, Claims, and Research Integrity

The Fieldworks portfolio contains material at very different levels of readiness. A credible organization cannot present every output in the same voice. The MindLAB therefore adopts a formal evidence ladder that travels with proposals, presentations, papers, demonstrations, and public exhibits.

Label	Meaning	Required support	Permitted language
Established context	A result accepted in the relevant literature	Primary or authoritative sources	"Established," "standard," with citation
Implemented	A function exists in code or apparatus	Version, path, demonstration, screenshots/logs	"Implemented," not "validated"
Measured	A protocol produced recorded observations	Methods, raw data, controls, uncertainty	"Measured in this instrument"
Replicated	Independent or repeated measurements agree	Predefined criteria and repeated datasets	"Replicated under stated conditions"
Simulated	A numerical model produced an output	Equations, parameters, code, sensitivity analysis	"Simulation predicts/suggests"
Proposed	An engineering or research design is specified	Requirements, roadmap, risks, validation gates	"Proposed," "candidate," "target"
Speculative	A broader interpretation	Explicit caveat and falsification path	"Hypothesis," "interpretive"

Label	Meaning	Required support	Permitted language
	exceeds current evidence		possibility"

The holonomy result

The strongest empirical thread in the current corpus is the measured closed-loop phase in a classical nonlinear instrument. Across six frozen, environment-zeroed sessions, the working paper reports orientation-odd behavior and approximate area-independence over the valid measurement range, with a small-area breakdown. The apparatus and result are explicitly classical. The phase is framed as a Hannay–Pancharatnam analog of Berry phase, not evidence that the synthesizer is a quantum computer or that quantum gravity has been observed.

Claims requiring revision or testing

- Quantum-language dashboards must be described as computational proxies or analog mappings unless a genuine quantum system is involved.
- Near-unity software variables such as a “Planck Gate” do not independently establish physical tunneling.
- Low numerical Lyapunov values in logged state variables do not by themselves establish quantum suppression of chaos.
- A field equation must pass dimensional, mathematical, and limiting-case review before physical conclusions are attached to it.
- CRE market, cost, resolution, and material-performance targets remain planning estimates until sourced and experimentally tested.
- Consciousness claims remain philosophical and model-building work unless tied to an appropriate empirical protocol.

Proposed research integrity practices

1. Assign every major claim an evidence label and source owner.
2. Archive source data, software commit, environment, and protocol for each result.
3. Pre-register the key comparison or rejection criterion for high-value experiments.
4. Use negative controls, parameter shuffles, and alternative models where applicable.
5. Invite an external method reviewer before publicizing strong physical interpretations.
6. Publish null results and rejected hypotheses when they materially define the framework.

06 Flagship Program: Cosmosis RDFT Suite

The public Cosmosis RDFT Suite is the organizational anchor because it shows that the framework is more than a collection of essays. The repository contains multiple runnable applications, supporting tools, examples, documentation, contribution guidance, and an MIT license. The project describes itself as an active experimental research suite with sharp edges and Mac-oriented launch workflows. This candid status should remain visible in all partnership material.

Why the suite matters

- It creates a common state vocabulary across audio, graphics, data bridges, topology views, and fabrication outputs.
- It allows the founder's conceptual framework to be inspected through actual behavior rather than prose alone.
- It can support reproducible examples and benchmark datasets that do not depend on accepting RDFT's strongest interpretations.
- Its visual and sonic outputs create an unusually strong public-engagement layer.
- Its modularity allows collaborators to work on one tractable component without inheriting the entire suite.

Twelve-month software hardening program

Workstream	Milestone	Evidence of completion
Release engineering	First versioned release with reproducible installers	Tagged release, checksums, clean-machine test
Documentation	Architecture map and project-specific quick starts	Published docs and newcomer usability test
Data schema	Shared state/log format with units and provenance	Schema, validators, sample datasets
Benchmarking	Reference runs for core instruments	Expected outputs, tolerances, regression checks
Analysis	Independent scripts for holonomy and topology measures	Tests against synthetic ground-truth data
Research packaging	Citable software and dataset release	DOI-bearing archive where appropriate
Accessibility	Keyboard, contrast, caption, and non-audio alternatives	Accessibility audit and issue log

Proposed collaboration packages

A university or laboratory partner does not need to adopt the entire suite. A computational mathematics group could audit the field equations. A visualization lab could validate manifold representations. A music technology program could study the instrument as a new interface for complex systems. A materials group could take only the Surface Foundry output and compare it with conventional topology optimization. A research software engineer could focus on releases, tests, and data provenance.

Canonical repository: https://github.com/FieldworksMindlab/Cosmosis_RDFT_Suite (public repository inspected July 17, 2026).

07 Flagship Program: Adaptive Fabrication and Surface Foundry

Surface Foundry is the most direct route from the Cosmosis field environment to physical evaluation. It generates meshes, reliefs, STL artifacts, material profiles, and process call sheets. The central

research question is practical: can the field-derived geometry deliver useful combinations of mass efficiency, stiffness, thermal behavior, acoustics, surface area, or manufacturability compared with appropriate baselines?

Terrestrial and off-world relevance

Large-format additive construction, robotic subtractive finishing, and in-situ resource utilization all need a design layer capable of adapting geometry to material and process constraints. The MindLAB's proposed contribution is not a new printer. It is a topology-generation and translation layer that can sit above multiple fabrication processes. That makes the work suitable for staged testing with existing equipment.

Minimum credible experiment

1. Define a baseline family: conventional solid, standard infill, and a recognized topology-optimized specimen.
2. Generate matched Surface Foundry specimens under the same envelope, material, and mass constraint.
3. Fabricate a statistically meaningful set using an accessible process.
4. Measure dimensional error, mass, stiffness, failure mode, print time, and post-processing demand.
5. Compare results with blinded geometry identifiers and publish both successes and failures.
6. Use the results to refine the material-profile adapter and fabrication call sheets.

Partnership targets

- University additive-manufacturing and robotics laboratories.
- Concrete printing, ceramic, polymer, or metal additive manufacturers.
- Architecture and engineering programs studying mass-optimized structures.
- NASA, Department of Defense, and civil-infrastructure programs where autonomous fabrication is mission-relevant.
- Museums and public laboratories able to exhibit the design-to-object process.

08 Flagship Program: Cathode Ray Epitaxy

Cathode Ray Epitaxy (CRE) is a high-risk engineering program exploring whether a scanned electron beam can act as a programmable energy field within a reactive precursor environment. The long-range vision is maskless, field-directed growth of electronic structures. The near-term research target is more modest and credible: model and demonstrate controlled beam-induced deposition or recrystallization whose morphology changes predictably with scan path, dose, thermal conditions, and reaction-diffusion parameters.

SCOPE DISCIPLINE True single-crystal epitaxy is a long-range stretch goal. Initial success means controlled deposition, predictable morphology, and useful material characterization. Public documents should not disclose novel process details that could compromise future patent options before counsel reviews the disclosure history.

Three-phase development path

Phase	Objective	Representative outputs	Working planning range
I — Simulation	Model coupled beam, precursor, thermal, diffusion, and growth dynamics	PDE model, design-space map, inverse path prototype, publication draft	\$80K–\$150K
II — Benchtop	Demonstrate controlled selective-area deposition	Modified instrument, dots/lines/patterns, characterization database	\$250K–\$400K
III — Device	Build multilayer structures and a simple functional device	Process design kit, functional test structure, scale-up plan	\$750K–\$1.5M

The current internal planning ask is approximately \$150,000 to complete Phase I simulation work, develop a defensible literature and prior-art review, and begin hardware planning. The broader three-year working range is approximately \$1.1–\$1.9 million. These are planning figures, not quotations or an approved budget; they must be rebuilt with partner facilities, indirect costs, personnel classifications, safety systems, and equipment access in view.

Immediate validation gates

- Independent prior-art and patent-landscape review covering EBID, e-beam-assisted CVD, beam annealing, direct-write electronics, and multibeam systems.
- Dimensional and numerical audit of the proposed coupled equations.
- A one-dimensional moving Gaussian source benchmark with known limiting behavior.
- Sensitivity analysis showing whether topology-aware path planning adds value beyond dose control.
- Materials and contamination review led by an experienced surface scientist.
- A safety and regulatory plan for vacuum, high voltage, radiation, reactive gases, and precursor handling.

09 Flagship Program: Voice of a Lyre

Voice of a Lyre proposes a biomimetic distributed sensing architecture in which many imperfect nodes cooperate through geometry, stochastic resonance, and coherent signal processing. The concept is motivated by the way biological systems assemble weak local signals into robust perception. Its most ambitious target is sub-classical gravitational field detection; its near-term value lies in the engineering of low-signal distributed arrays.

Researchable core

A strong first program does not require a gravitational claim. It can ask whether controlled noise injection, array geometry, phase alignment, and adaptive estimation improve detection of a known weak signal under realistic drift and calibration error. That question is useful in structural monitoring, acoustics, photonics, magnetometry, environmental sensing, and distributed instrumentation.

Prototype ladder

Stage	System	Question
Simulation	Synthetic heterogeneous sensor array	When does stochastic resonance improve detectability?
Bench array	8–32 inexpensive nodes with injected reference signal	Can calibration and coupling recover a signal below individual-node thresholds?
Environmental test	Distributed acoustic, vibration, magnetic, or optical nodes	Does the method retain value under real drift and interference?
Specialized partnership	Higher-sensitivity transducers	Is there a credible path toward the original field-sensing ambition?

The Voice of a Lyre name also provides a public narrative: a lyre does not create meaning from one string in isolation, but from relation, tuning, resonance, and listening. That makes the project especially suitable for a combined engineering, sound, and public-learning program.

10 Consciousness, Arts, and Public Inquiry

The MindLAB's consciousness work asks whether experience is better described as a stable relational organization than as an isolated substance. The post-Cartesian white paper connects embodied cognition, predictive processing, information, nonlinear dynamics, resonance, and field metaphors. It is explicitly a speculative synthesis rather than a completed physical theory.

Why this belongs in the organization

Consciousness research gives the portfolio a human horizon. The instruments do not only calculate; they produce experiences of pattern, interference, timing, coherence, and breakdown. Those experiences can motivate empirical questions about perception and learning, but they can also stand as works of media art and public philosophy. The organization should support both uses while maintaining a boundary between an aesthetic experience and evidence about the nature of consciousness.

Public MindLAB program

- Interactive demonstrations of feedback, attractors, phase, resonance, and topology.
- Sound-and-image installations driven by live environmental data.
- Workshops in creative coding, fabrication literacy, scientific visualization, and responsible speculation.

- Partnerships with schools, libraries, museums, arts organizations, and community colleges.
- Accessible materials that give participants nonvisual and nonauditory routes into the same concepts.
- Evaluation of what participants learn, question, build, and retain.

Funding fit and organizational constraints

Arts funding generally requires an eligible nonprofit, public entity, tribe, or qualified partner. For example, the National Endowment for the Arts Grants for Arts Projects program supports project-based arts activity but does not fund individuals or commercial enterprises directly and imposes organizational history and cost-share requirements. Fieldworks MindLAB should therefore pursue arts funding through an eligible institutional partner until its own status and programming history meet the relevant requirements.

Funding eligibility and deadlines change. Current program information must be reverified at arts.gov and the relevant agency site for each application cycle.

11 Partnership and Business Model

Fieldworks MindLAB is best organized as a portfolio platform with shared technical infrastructure and multiple revenue or support channels. Grant funding alone should not be expected to sustain every activity, and commercial pressure should not force speculative research to masquerade as a finished product.

Partnership pathways

Partner	What the MindLAB contributes	What the partner contributes	Typical output
University laboratory	Framework, code, instruments, research questions	Methods expertise, facilities, students, review	Joint study, dataset, paper, prototype
National laboratory	Unconventional concepts and integration platform	Specialized infrastructure and mission context	Feasibility evaluation or cooperative R&D
Industry R&D	Rapid concept generation, software, prototypes	Requirements, materials, equipment, commercialization	Sponsored research or licensed tool
Museum/arts organization	Interactive instruments and narrative	Audience, curation, education, production	Exhibition, workshop, public dataset
School/community partner	STEAM modules and creative technology	Learners, educators, local context	Pilot curriculum and evaluation
Funder/philanthropy	Integrated high-risk research portfolio	Flexible capital and convening power	Milestone program and public reporting

Candidate support model

- Sponsored R&D agreements for defined technical work.
- Federal SBIR/STTR or mission-agency proposals where the applicant and topic match current requirements.

- Subawards or consultant roles on university-led proposals.
- Software customization, research instrumentation, and visualization services.
- Licensing of protected process, control, or design technology.
- Workshops, installations, performances, and educational programming.
- Philanthropic seed support for open research assets and public access.

A practical two-entity future

If the portfolio grows, a dual structure may be appropriate. A company can hold commercial contracts, IP, equipment, and product development. A nonprofit affiliate or long-term fiscal partnership can support education, exhibitions, open datasets, and community access. The exact structure requires legal, tax, grant, and conflict-of-interest advice; this prospectus recommends the functional separation without prescribing a legal conclusion.

12 Three-Year Roadmap and Funding Strategy

Program roadmap

Period	Organizational objective	Technical objective	Partnership objective
0–6 months	Establish entity strategy, advisory circle, source register, and grant calendar	Release engineering, benchmark datasets, experiment protocols	Secure method reviewers and facility conversations
6–12 months	Publish prospectus derivatives and operating policies	Versioned Cosmosis release; first independent holonomy replication plan; CRE model	Submit first collaborative proposals and sponsored scopes
Year 2	Add technical staff or fellows and formal project accounting	Fabrication specimen study; Voice of a Lyre bench array; CRE deposition proof	Active university, industry, and public-program partnerships
Year 3	Demonstrate a sustainable mixed support model	Validated prototypes, peer-reviewed outputs, process documentation	Multi-institution program and scale-up decision

Funding ladder

The funding strategy should match readiness. Small unrestricted or philanthropic support can fund documentation, release engineering, and pilot studies. University partnerships can support formal review and shared facilities. Federal small-business programs can support high-risk technical development when the company, topic, and commercialization plan are eligible. Larger awards should follow—not precede—clear bench evidence and a capable team.

Tier	Indicative need	Best use	Decision gate
Seed	\$25K–\$75K	Documentation, release, data cleanup, advisory review	A reproducible public technical baseline
Focused feasibility	\$75K–\$200K	CRE simulation, array prototype, fabrication	Evidence that one

Tier	Indicative need	Best use	Decision gate
		coupons	flagship merits scale-up
Phase I / collaborative grant	\$150K–\$350K	Defined R&D scope with commercialization or research plan	Partnered prototype and validated milestones
Phase II / program build	\$500K–\$2M+	Equipment, personnel, multi-year prototype development	Demonstrated feasibility and committed partners

Initial use-of-funds concept

Category	Share	Purpose
Research and engineering labor	40–50%	Software hardening, modeling, protocol design, analysis
Facilities and equipment access	15–25%	Fabrication, microscopy, vacuum, sensors, characterization
Prototype materials and safety	10–15%	Components, consumables, controls, compliance
External review and professional services	8–12%	Methods review, IP counsel, entity/grant advice
Documentation and public dissemination	8–12%	Technical writing, data publication, accessibility, demonstrations
Contingency	5–10%	Unknowns inherent in early-stage research

These percentages are planning guidance. A grant-ready budget must be rebuilt against the chosen program rules, partner indirect costs, salary basis, equipment classification, subawards, cost share, and the actual scope of work.

13 Governance, IP, Ethics, and Risk

Intellectual-property boundary

The Cosmosis RDFT Suite is publicly available under the MIT license. That supports adoption and collaboration but also means code already released under that license cannot later be made proprietary for downstream users. New patentable processes, hardware, precursor protocols, control methods, datasets, trademarks, and unreleased implementation details may be governed separately. The organization needs a written IP map before further public disclosure of CRE or other potentially patentable work.

Recommended governance

- A small scientific and engineering advisory circle with no obligation to endorse the framework.
- Written contributor, authorship, data, and IP agreements before substantive collaboration.
- A conflict-of-interest register for reviewers, partners, and sponsored research.

- Milestone reviews that can stop, narrow, or redirect projects.
- Separate public, partner-confidential, and patent-counsel review versions of sensitive material.
- A corrections policy for public documents and software claims.

Risk register

Risk	Consequence	Mitigation
Overclaiming physical significance	Loss of scientific credibility	Evidence labels, external review, narrow language, publish controls
Portfolio sprawl	Insufficient depth in any program	Quarterly priority gates and one flagship experiment at a time
Founder bottleneck	Slow delivery and fragile continuity	Documentation, modular ownership, advisors, funded technical roles
Patent/publication conflict	Loss of rights or blocked collaboration	Pre-disclosure IP review and explicit information tiers
Facility and safety barriers	CRE or fabrication work cannot proceed	Partner with established laboratories; phase work through simulation
Funding mismatch	Ineligible or uncompetitive applications	Live eligibility checks and partner-led submissions where appropriate
Software fragility	Demonstrations cannot be reproduced	Versioned releases, clean-machine tests, regression datasets
Public misunderstanding	Metaphor mistaken for settled science	Audience-specific framing and visible epistemic labels

14 Reference Framework and Development Lineage

Fieldworks MindLAB developed through active dialogue with established and emerging work in topology, driven systems, relational time, energy-based memory, and gauge-field geometry. The references below do not “prove” RDFT. They identify the technical ideas that shaped the vocabulary, interfaces, and experimental questions.

Development reference	Contribution to the MindLAB lineage	How it will be used
Wassner et al. — Holonomic quantum computation	Adiabatic control, non-Abelian holonomy, robustness of geometric operations	Context for loop protocols; quantum/classical distinction retained
Castro et al. — Floquet engineering	Driven systems and optimal control of Floquet behavior	Context for drive design, quasienergy, and control objectives
Barontini — Problem of time with cold atoms	Relational time ordered by internal/entropic variables	Context for time-as-observable discussions and experimental discipline
Schmidt — Topological insulators and invariants	Berry phase, curvature, Chern number, robust boundary behavior	Introductory topology and invariant vocabulary
Ramsauer et al. — Hopfield networks	Energy-based associative memory and equivalence to attention	BasinNET, attractor, and memory research framing
Nielsen — Complex Hopf gauge-gravity preprint	Nested Hopf shells, Beltrami spectrum, holonomy, torsion, unification claims	Major comparative framework, explicitly labeled preprint

Reference policy

- Third-party papers are cited and summarized; they are not reproduced in the authored appendices.
- Brian Fahey's complete source papers are preserved in Volume II with their original version and claim language.
- The prospectus uses its own evidence labels even when an appendix uses stronger historical wording.
- A consolidated bibliography accompanies every substantial derivative packet.
- Preprints and student reviews are identified by publication status rather than presented as peer-reviewed consensus.

15 Invitation to Partner

Fieldworks MindLAB is looking for partners who recognize value in unusual synthesis and also insist on disciplined proof. The portfolio is broad, but each proposed collaboration can be narrow: review one equation, replicate one measurement, test one geometry family, simulate one deposition process, build one array, or host one public demonstration.

Priority partnership conversations

- Applied mathematics and topology researchers interested in rigorous review of the framework and holonomy protocols.
- Research software engineers and visualization specialists able to harden the Cosmosis suite.
- Materials, surface-science, microscopy, and nanofabrication groups willing to scope CRE feasibility.
- Additive-manufacturing and robotics laboratories able to test Surface Foundry geometries.
- Sensor, photonics, acoustics, or controls groups interested in the Voice of a Lyre array architecture.
- Museums, arts organizations, schools, and libraries interested in a public MindLAB pilot.
- Funders prepared to support high-risk research through transparent milestones and independent evaluation.

A PRACTICAL FIRST MEETING Select one flagship program, identify the partner's real facility or mission need, define a 90-day proof with a rejection criterion, and agree on ownership, safety, publication, and data terms before work begins.

Brian Fahey · Founder and Principal Investigator · Fieldworks MindLAB · Ocala, Florida
Repository: https://github.com/FieldworksMindlab/Cosmosis_RDFT_Suite
LinkedIn: <https://www.linkedin.com/in/brianpfahey/>

Appendix A Source and Appendix Register

ID	Full-length Volume II source	Author	Date/version	Editorial status
A	Toward a Post-Cartesian Science of Consciousness	Brian Fahey	2025–2026	Speculative theoretical synthesis
B	A Measured Topological Geometric Phase in a Driven Nonlinear Field Instrument	Brian Fahey	Working draft v0.2	Classical measurement paper
C	RDFT Master Guide	Brian Fahey	2026	Comprehensive founder framework
D	Adaptive Fabrication Systems for Terrestrial and Off-World Construction	Brian Fahey / Fieldworks MindLAB	July 2026	Federal R&D brief
E	Cathode Ray Epitaxy	Brian Fahey / Fieldworks MindLAB	2026-07-17	Proprietary engineering brief
F	Nonlinear Phi Field Equation: Derivatives and Extensions	Brian Fahey	2026	Compact mathematical working reference
G	Voice of a Lyre	Brian Fahey	2026	Conceptual engineering white paper
H	Where Materials, Physics, and Sound Become a Public Instrument	Brian Fahey / Fieldworks MindLAB	July 2026	STEAM and arts brief
I	Resolution, Recursion, and Reality	Brian Fahey	2026	Conceptual paper
J	Resolution, Recursion, and Reality: Preprint	Brian Fahey	2026	Formalized preprint version
K	Personal Statement	Brian Fahey	2026-07-07	Founder's autobiographical and academic statement
L	Floquet Systems and Adiabatic Processes Under Recursive Differentiation	Brian Fahey (with AI assistance)	Version 1.0 — 2026-05-21	Internal working framework with implementation code

The complete, unabridged versions are compiled in Volume II. The appendix title sheets are new editorial matter; the underlying papers remain in their source form.

Appendix B Selected References

Barontini, Giovanni. 2026. "Testing the Problem of Time with Cold Atoms." *Physical Review Research* 8: L022047. <https://doi.org/10.1103/1h9j-df4k>.

Berry, Michael V. 1984. "Quantal Phase Factors Accompanying Adiabatic Changes." *Proceedings of the Royal Society A* 392: 45–57. <https://doi.org/10.1098/rspa.1984.0023>.

Castro, Alberto, Umberto De Giovannini, Shunsuke A. Sato, Hannes Hübener, and Angel Rubio. 2023. "Floquet Engineering with Quantum Optimal Control Theory." *New Journal of Physics* 25: 043023. <https://doi.org/10.1088/1367-2630/accb05>.

- Nielsen, Jennifer Lorraine. 2026. "The Complex Hopf Fibration as the Canonical Space for Gauge–Gravity Unification: The Field, Universal Action, and Particle Spectrum." Preprints.org, Version 4. <https://doi.org/10.20944/preprints202604.0315.v4>.
- Ramsauer, Hubert, et al. 2021. "Hopfield Networks Is All You Need." arXiv:2008.02217.
- Schmidt, Christian. 2021. "Topological Insulators and Their Invariants." Department of Physics, University of British Columbia, introductory review.
- Simon, Barry. 1983. "Holonomy, the Quantum Adiabatic Theorem, and Berry's Phase." *Physical Review Letters* 51: 2167–2170. <https://doi.org/10.1103/PhysRevLett.51.2167>.
- Wassner, Clara, Tommaso Guaita, Jens Eisert, and Jose Carrasco. 2026. "Holonomic Quantum Computation: A Scalable Adiabatic Architecture." *Quantum*. arXiv:2502.17188.
- National Endowment for the Arts. 2026. "Grants for Arts Projects." <https://www.arts.gov/grants/grants-for-arts-projects>.
- U.S. Small Business Administration. 2026. "America's Seed Fund: SBIR/STTR." <https://www.sbir.gov/about>.
- Fieldworks MindLAB. 2026. "Cosmosis RDFT Suite." https://github.com/FieldworksMindlab/Cosmosis_RDFT_Suite.

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